

TANGSHAN, China

WMO: 545340

Lat: 39.5816N

Lon: 118.0931E

Elev: 29

StdP: 100.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.6	-26.6	0.3	-5.8	-24.5	0.4	-6.1	9.3	-1.9	8.1	-3.2	1.5	320	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	33.8	23.2	32.5	23.2	31.3	22.9	27.1	30.8	26.1	29.7	25.3	28.7	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.6	25.1	20.3	28.6	24.3	19.3	27.8	85.9	31.0	81.8	29.8	78.1	28.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
7.7	6.4	5.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-16.0	36.4	3.6	1.7	-18.5	37.6	-20.6	38.6	-22.6	39.6	-25.2	40.8		
			-16.9	28.2	3.1	0.8	-19.2	28.8	-21.0	29.3	-22.8	29.8	-25.0	30.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-4.1	-0.5	6.5	13.9	20.0	24.3	26.8	25.6	21.0	13.6	4.9	-2.0		
	DBStd	11.32	3.53	3.80	4.10	3.85	3.23	2.69	2.20	2.19	3.10	3.80	4.22	3.38		
	HDD10.0	1412	438	293	125	10	0	0	0	0	0	12	160	373		
	HDD18.3	2951	697	526	368	141	20	1	0	0	12	152	404	632		
	CDD10.0	2343	0	0	15	126	309	428	519	484	331	124	6	0		
	CDD18.3	841	0	0	0	7	71	179	261	226	93	5	0	0		
	CDH23.3	7688	0	0	2	71	649	1613	2648	2056	619	31	0	0		
	CDH26.7	2719	0	0	0	9	188	621	1053	702	143	3	0	0		
(14) Wind	WSAvg	2.4	2.2	2.4	2.9	3.1	3.0	2.6	2.3	2.0	2.0	2.1	2.1	2.1		
(15) Precipitation	PrecAvg	635	4	6	10	25	38	83	195	182	57	22	8	4		
	PrecMax	1009	25	31	38	91	108	197	397	350	167	69	32	25		
	PrecMin	386	0	0	0	1	5	17	27	49	3	0	0	0		
	PrecStd	179	6	7	9	24	25	49	91	87	39	18	8	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	12.4	21.9	27.7	33.2	35.8	36.2	34.0	31.3	26.5	17.8	8.6		
		MCWB	1.0	4.8	11.5	15.4	18.7	20.3	23.4	25.6	21.3	16.9	10.0	2.7		
	2%	DB	4.8	9.9	18.8	25.3	30.7	33.4	34.2	32.4	29.7	23.7	15.6	6.4		
		MCWB	-0.5	3.4	9.6	14.5	17.8	21.0	24.4	24.7	20.3	15.6	9.1	1.9		
	5%	DB	3.1	8.0	16.0	23.3	28.8	31.8	32.7	31.2	28.3	21.7	13.4	4.7		
		MCWB	-1.5	2.4	7.9	13.2	17.3	20.8	24.6	24.2	20.0	14.5	8.1	0.6		
	10%	DB	1.5	6.1	13.6	21.2	27.1	30.2	31.4	30.2	26.9	19.9	11.4	3.2		
		MCWB	-2.2	1.4	6.2	12.2	16.5	20.5	24.3	23.8	19.3	13.4	6.9	-0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	6.0	13.2	18.1	21.7	24.8	28.2	28.1	25.0	19.2	12.7	4.9		
		MCDB	5.5	9.9	20.0	25.3	30.0	30.6	32.4	31.5	28.0	23.2	15.2	6.8		
	2%	WB	0.1	4.4	10.5	16.1	20.2	23.6	27.2	27.1	23.0	17.8	10.4	2.6		
		MCDB	3.4	8.3	16.7	23.0	27.5	28.8	31.4	30.4	26.8	21.3	13.5	5.0		
	5%	WB	-0.8	3.0	8.6	14.7	19.1	22.7	26.5	26.1	21.9	16.4	9.0	1.4		
		MCDB	2.5	7.3	14.6	20.9	25.7	28.0	30.4	29.3	25.4	19.8	12.5	3.9		
	10%	WB	-1.8	1.6	6.9	13.3	18.0	21.9	25.7	25.3	20.9	15.1	7.5	0.2		
		MCDB	1.3	5.7	12.9	19.3	24.5	27.4	29.3	28.4	24.7	18.4	10.9	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.0	10.7	11.1	11.4	9.6	7.7	8.1	9.9	10.5	9.4	9.0		
		MCDBR	11.4	12.1	13.9	14.0	14.6	12.6	9.8	9.3	11.3	12.5	11.8	10.9		
	5% WB	MCWBR	7.5	7.4	7.1	6.2	5.4	4.1	3.4	3.6	4.1	5.5	6.4	7.0		
		MCWBR	10.3	11.0	12.9	11.7	11.6	9.2	7.7	7.4	8.6	9.8	10.6	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.439	0.519	0.545	0.621	0.666	0.789	0.792	0.729	0.655	0.631	0.508	0.457		
	taud		1.922	1.779	1.750	1.649	1.600	1.448	1.482	1.576	1.650	1.642	1.832	1.923		
	Ebn at Noon		669	664	704	683	665	587	579	601	609	556	587	610		
	Edh at Noon		137	182	207	244	262	305	292	260	227	207	150	128		
(44) All-Sky Solar Radiation	RadAvg		2.38	3.12	4.49	5.35	6.00	5.32	4.79	4.70	4.22	3.22	2.42	2.08		
	RadStd		0.23	0.33	0.39	0.27	0.37	0.51	0.38	0.28	0.31	0.29	0.30	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	-1.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63	+46		

Nomenclature: See separate page